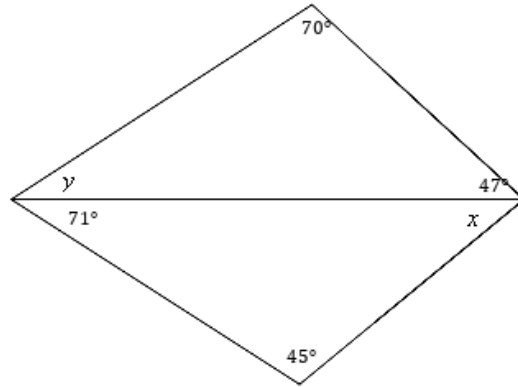


Angles of polygons

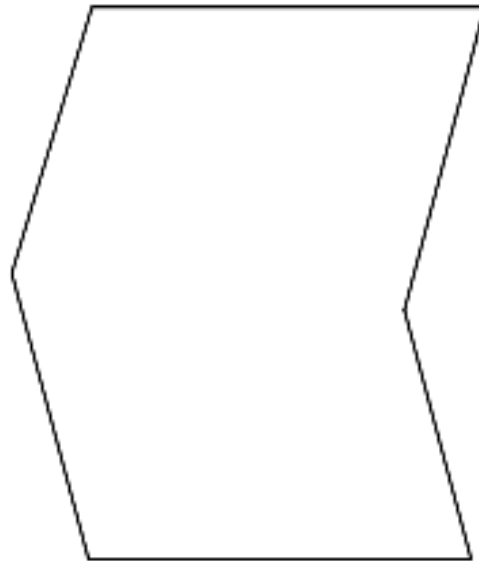


- 1 State the sum of the interior angles in a quadrilateral.
- 2 State the sum of exterior angles for any polygon.
- 3 Consider the quadrilateral in the adjacent diagram.

- a Find the value of the angle marked y .
- b Find the value of the angle marked x .



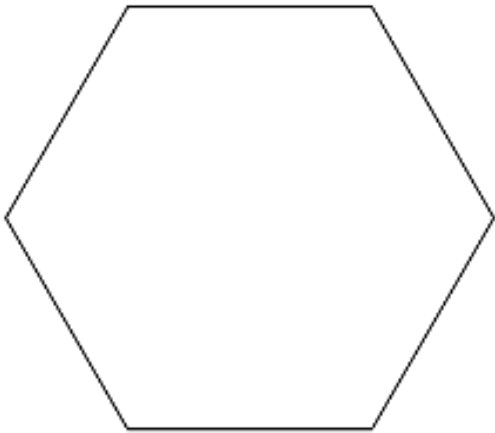
- 4 Consider the non-convex hexagon in the diagram shown.
- a Find the least number of triangles that it can be divided into.
 - b Determine the interior angle sum of the non-convex hexagon.



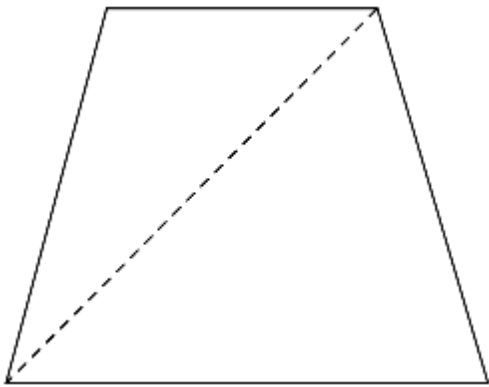
5 Consider the convex hexagon in the adjacent diagram.



- a Find the least number of triangles that the hexagon can be divided into.
- b Determine the interior angle sum of the hexagon.



6 Consider the adjacent diagram.
The quadrilateral is divided into two triangles (this is the least number of triangles that a quadrilateral can be divided into). The interior angle sum of a quadrilateral is therefore equal to double the interior angle sum of a triangle.
Complete the following table.



Name of the polygon	Number of sides	Least number of triangles	Interior angle sum
Pentagon			
Hexagon			
Heptagon			
Octagon			
Nonagon			
Decagon			

7 Consider the pentagon to the right.



a Find the value of the following:

i $a + b + c$

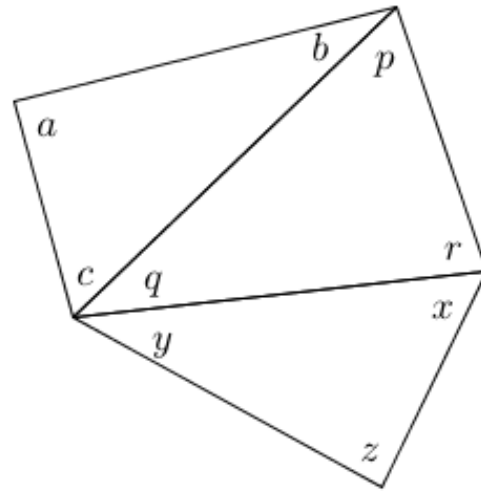
ii $p + q + r$

iii $x + y + z$

b Determine the angle sum of the pentagon.

c How many non-overlapping triangles can an n -sided figure be split into?

d Hence or otherwise, state the interior angle sum of an n -sided polygon.



8 Find the interior angle sum of a regular polygon whose exterior angles measure 40° .

9 Consider the adjacent quadrilateral.

a Find the value of the following angles:

i x

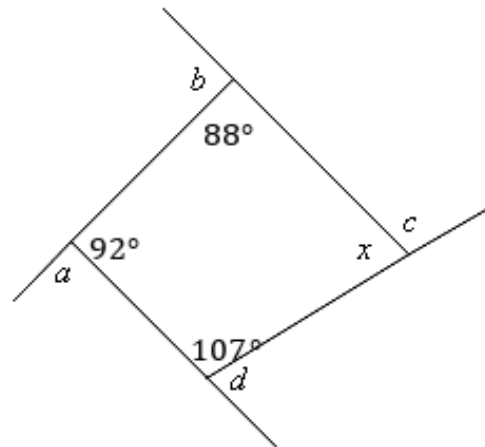
ii a

iii b

iv c

v d

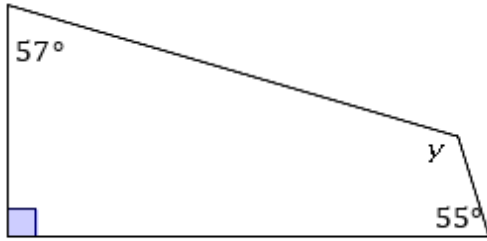
b State the sum of exterior angles in a quadrilateral.



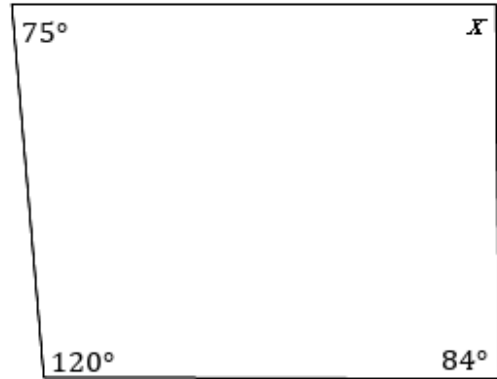
10 Find the value of the variable(s) in the following diagrams.



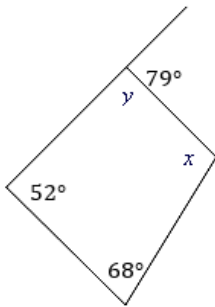
a



b

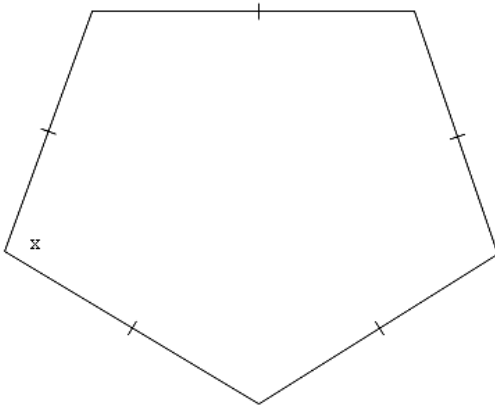


c

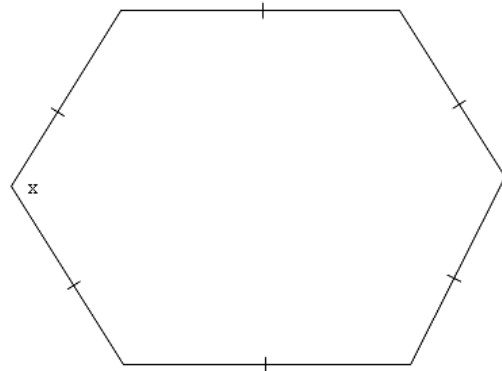


11 Given that each of the following polygon is regular, find the value of x .

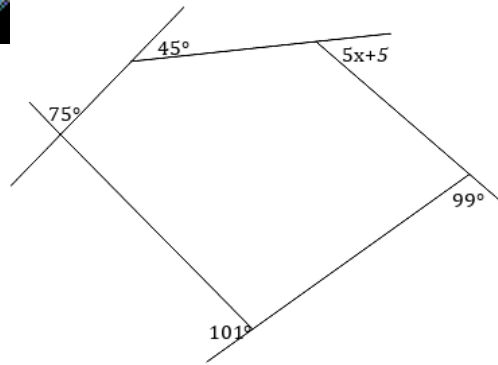
a



b



- 12 Consider the following diagram.
Calculate x .



- 13 The interior angles of an n -sided regular polygon each have a measure of x° .
Write down a formula for x in terms of n .

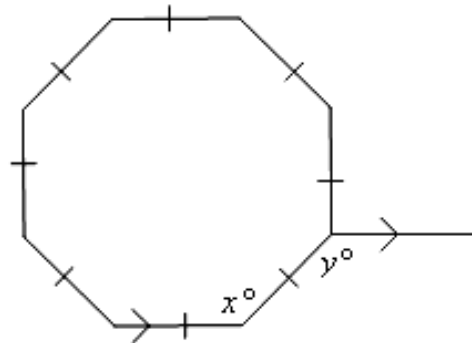
- 14 Find the measure of each of the interior angles in a regular:

- | | |
|-------------------|-------------|
| a 6-sided polygon | b Octagon |
| c Decagon | d Dodecagon |

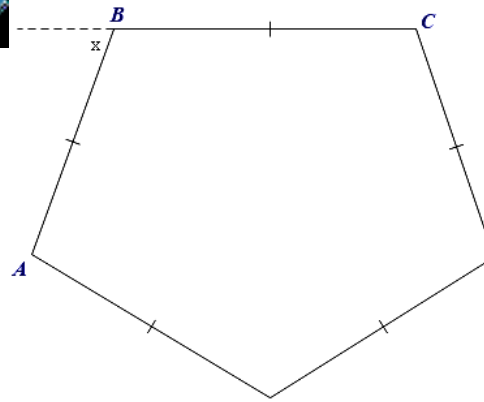
- 15 Find the measure of each exterior angle of a regular polygon with 20 sides.

- 16 Consider the regular polygon in the adjacent diagram.

- a Solve for x .
b Solve for y .

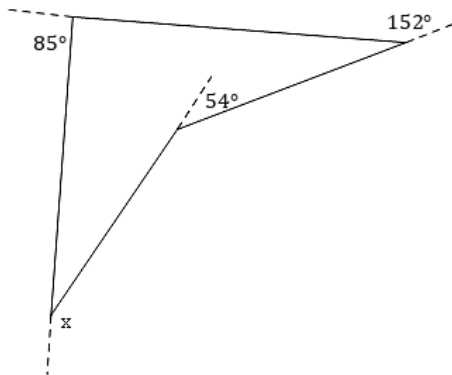


- 17 Given that the shape in the diagram is a regular pentagon, find the value of x .

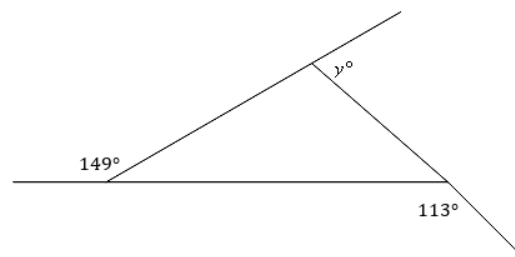


- 18 Find the value of the variable in the following diagrams.

a



b



- 19 The exterior angles of an n -sided regular polygon each have a measure of x° . Write down a formula for x .

- 20 Given the following angle measures of a regular polygon:

- i Find the number of sides of the polygon.
- ii Identify the shape of the polygon if it exists.

a Interior angles equal to 150° each.

c Interior angles equal to 120° each.

e Exterior angles equal to 72° each.

g Exterior angles equal to 75° each.

i Interior angles equal to 100° each.

b Interior angles equal to 140° each.

d Exterior angles equal to 36° each.

f Exterior angles equal to 45° each.

h Exterior angles equal to 50° each.

j Interior angles equal to 130° each.

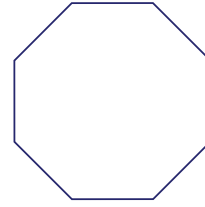


Additional questions

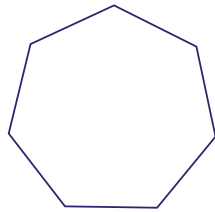
21 Find the measure of an interior angle in each of the following regular polygons:

a Hexagon

b



c

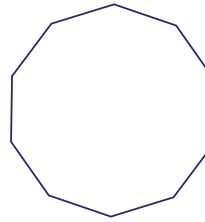


d 22-gon

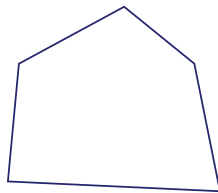
22 Find the sum of the interior angles in each of the following polygons:

a Decagon

b



c



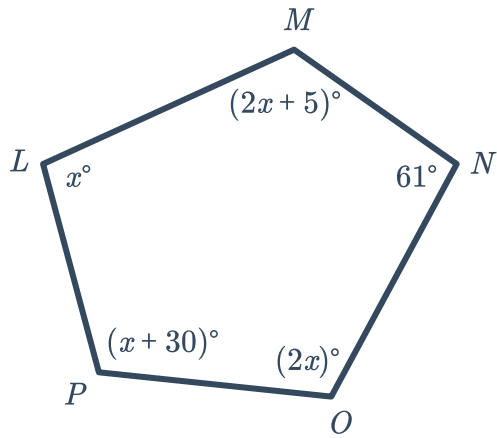
d 22-gon

23 What is the sum of the exterior angles of a 13-gon?

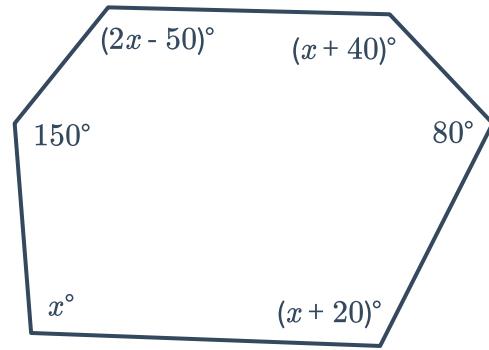
24 Solve for x in the following polygons:



a

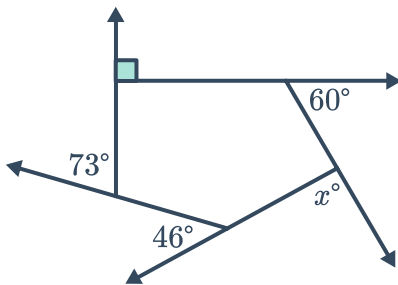


b

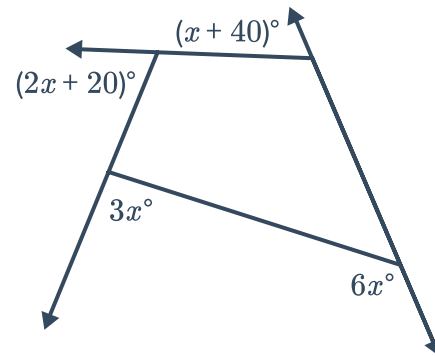


25

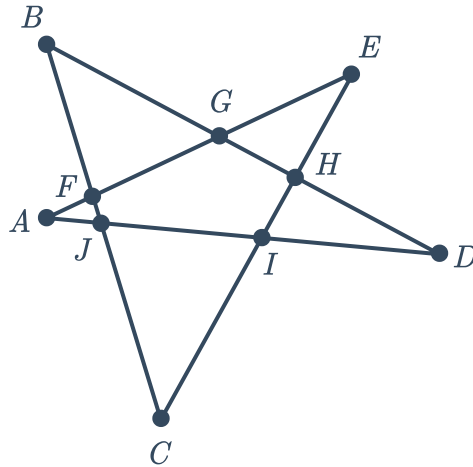
a Solve for x .



b

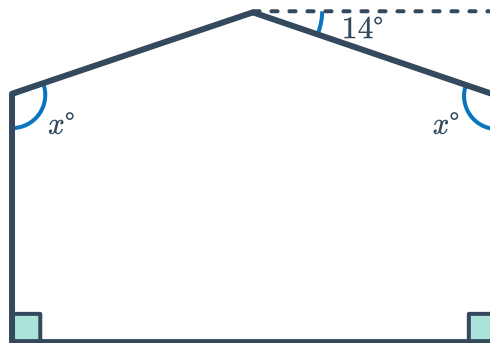


- 26 In the diagram, $m\angle B = 43^\circ$, $m\angle FJI = 109^\circ$, $m\angle GHI = 92^\circ$, $m\angle JFG = 101^\circ$, and $m\angle C = 45^\circ$.



Solve for the following:

- $m\angle A$
 - $m\angle FGH$
 - $m\angle HIJ$
 - $m\angle BFG + m\angle EGH + m\angle DHI + m\angle CIJ + m\angle AJF$
- 27 A house is built with a pitched roof. The engineer states that the angle of the pitch is 14° . Solve for x .

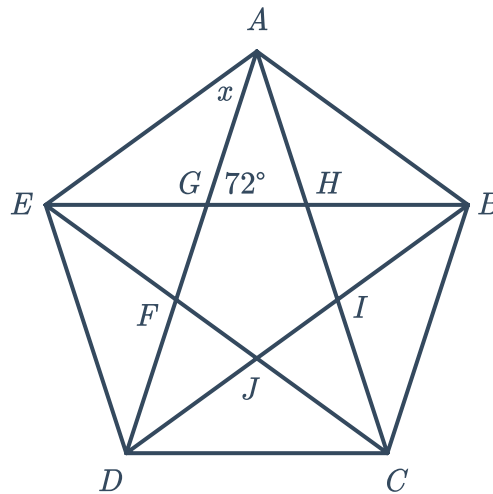


28 Determine the number of sides of a polygon in:



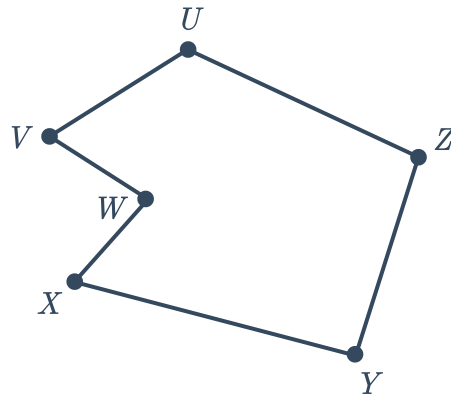
- a The interior angles sum to 360°
- b Each interior angle has a measure of 135°
- c Each exterior angle has a measure of 24°
- d Each exterior angle has a measure of 15°
- e The interior angles sum to 1980°
- f Each interior angles has a measure of 160°

29 $ABCDE$ is a regular pentagon.



- a What is the measure of $\angle AED$? Explain how you know.
- b Explain why $x = 36^\circ$.

30 Given: Hexagon $UVWXYZ$

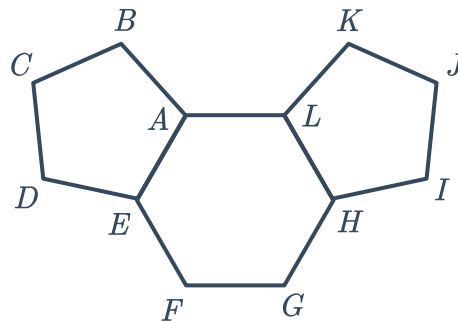


- a Divide the hexagon into six triangles, all of which share a common vertex T on the interior of $UVWXYZ$.
- b Show that the sum of the angles in the six triangles is 1080° .
- c Show that the sum of the six angles that meet at T is 360° .
- d Using the results from parts a) and b), show that the sum of the interior angles of Hexagon $UVWXYZ$ is 720° .

- 31 The pattern on a soccer ball is a tessellation  repeated pattern with no overlaps or gaps. On a three-dimensional soccer ball, the tessellation consists of regular pentagons and hexagons:



When laid flat, a piece of the pattern looks like this:



Explain why the pattern on a soccer ball is impossible to tessellate on a two-dimensional surface.